

Prem Kumar Shaw

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PROFESSIONAL SUMMARY

B.Tech Computer Science (Data Science) graduate with hands-on experience building backend services, REST APIs, and automation utilities using Python and FastAPI. Strong foundation in object-oriented programming, data structures, SQL/NoSQL databases, Git workflows, and software testing. Internship experience includes large-scale ETL pipelines, log/data processing, and dashboard development. Passionate about backend engineering, automation, and AI-powered software systems.

TECHNICAL SKILLS

Programming Languages : Python, SQL, JavaScript, C++
Backend & APIs : FastAPI, REST APIs, Pydantic, SQLAlchemy (ORM), JWT Authentication, API Integration
Databases : MongoDB (NoSQL), MySQL (SQL)
DevOps & Tools : Git, GitHub, Docker, CI/CD basics, VS Code, Jupyter Notebook, Google Colab, Power BI
Data Analytics : data processing, ETL, dashboards, PySpark, Streamlit, Pandas, NumPy, Scikit-learn
AI Tools : ChatGPT, Claude, Gemini API, AI-assisted coding
Core Concepts : DSA, OOP, DBMS, SDLC, Operating Systems, Computer Networks, Basic System Design, Debugging & Unit Testing

INTERNSHIP EXPERIENCE

Forcas Studio Limited — Data Analytics Intern

Kolkata | Aug 2025 – Oct 2025

- Managed end-to-end data operations for Amazon, Flipkart, Ajoio, and Meesho e-commerce platforms.
- Automated daily data processing of 800,000+ records using Python, improving pipeline efficiency by 50%.
- Designed and executed ETL operations and advanced SQL queries to derive actionable business insights.
- Built interactive Excel dashboards and Pivot Tables to support data-driven business decisions.

Techno Exponent — Machine Learning Engineer Intern

Remote | Aug 2025 – Nov 2025

- Built and optimized machine learning models for structured datasets using Python and Scikit-learn.
- Developed a CatBoost regression model with advanced feature engineering to predict diamond prices, achieving $R^2 \approx 0.97$.
- Developed backend logic and deployed the ML model using Streamlit for real-time predictions.

PROJECTS

AI Resume Interview Coach (In Progress) | Python, FastAPI, MongoDB, Gemini API, JWT Authentication

- Developing an AI-powered mock interview platform that generates personalized interview questions from uploaded resumes.
- Building REST APIs with FastAPI and integrating MongoDB for user data and interview history.
- Implementing AI-powered answer evaluation and personalized feedback using the Gemini API.
- Integrating Speech-to-Text and Text-to-Speech for voice-based interview interactions.

Smart Expense Tracker API | Python, FastAPI, Pydantic, Pytest, REST APIs | [GitHub](#)

- Designed and developed a modular RESTful Expense Tracker API using FastAPI, implementing CRUD operations for expense management.
- Designed REST endpoints for expense creation, retrieval, filtering, summary generation, and deletion.
- Implemented request validation using Pydantic and structured JSON-based data persistence.
- Wrote automated API test cases using Pytest to verify endpoint functionality and improve reliability.
- Documented APIs using Swagger/OpenAPI, enabling interactive testing and developer-friendly documentation.

EDUCATION

Brainware University — B.Tech in Computer Science & Engineering (Data Science)

Kolkata, India

CGPA: 8.57 / 10 | Percentage: 82%

2022 – 2026

CERTIFICATIONS

- Software Engineer Intern Certificate — [HackerRank](#)
- SQL (Basic & Intermediate) Certifications — [HackerRank](#)
- Artificial Intelligence — [IBM SkillsBuild](#)
- Python Certificate — [Udemy](#)

ACHIEVEMENTS

- Solved 180+ coding problems on LeetCode and HackerRank, covering Data Structures, Algorithms, and SQL.
- Solved 50+ SQL problems on LeetCode and earned SQL (Basic & Intermediate) badges.
- School Topper (1st Rank) in Class X (2019) and Class XII (2021).